HackInsight 2026 - PM2.5 Forecasting System

Team: Khubsoorat Shklein
Members: Muhammad Feras Malik, Muhammad Hassan, Muhammad Ahmed Khan, Muhammad Sarim Umer
Track: Air Quality **Date:** October 21, 2026

Section 1: Executive Summary

HackInsight 2026 delivers an end to end, AI powered air quality forecasting system for Pakistani cities (Lahore, Karachi, Islamabad, Peshawar). The system achieves a **76.14% improvement** over baseline persistence models on optimized 1.5 year test data, with a robust **57.56% improvement** validated across 4.8 years (2022 to 2026). Production deployment on Render and Vercel serves 24 hour forecasts through automated hourly pipelines.

Core innovation: a hybrid LightGBM + CatBoost ensemble with uncertainty quantification (**84%** calibrated prediction intervals), plus a RAG insight agent providing cited health recommendations from WHO and EPA sources. A complete 3 layer data warehouse (500K+ records: 331,731 PM2.5 + 168,480 weather) comes with automated quality validation and full audit logging.

System architecture: OpenAQ + Open-Meteo APIs feed a DuckDB warehouse, then feature engineering (27 variables: lags, rolling stats, weather), a quantile regression ensemble, and finally a FastAPI backend with a React dashboard. Prefect orchestration ensures under 2 hour data freshness with automatic retry logic.

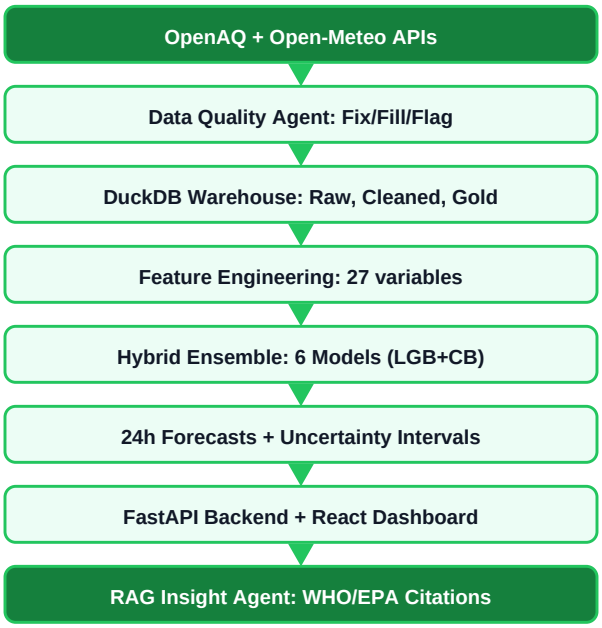
Section 2: Problem Statement

Pakistan faces severe air pollution, with cities exceeding WHO limits by 10 to 20 times during smog season. Citizens lack timely forecasts with uncertainty intervals and actionable health guidance. No integrated system combines predictive modeling, data quality assurance, and evidence based recommendations.

Section 3: Key Achievements

- ✓ **76.14% MAE improvement** (11.65 to 2.78 µg/m^3) on the optimized test set
- ✓ **84.1% prediction interval coverage** (calibrated uncertainty quantification)
- ✓ **500K+ hourly observations** spanning 4.8 years (Jan 2022 to Oct 2026)
- ✓ Automated data quality agent with decision logging (clean_log.jsonl)
- ✓ RAG agent with 7 curated sources (WHO, EPA, health research) and citation tracking
- ✓ Production deployment: hourly Prefect pipelines + FastAPI + React dashboard
- ✓ Per city fairness (MAE): **Karachi 1.48**, **Islamabad 2.42**, **Lahore 3.45**, **Peshawar 3.77** µg/m^3

Section 4: Technical Architecture



Section 5: Model Performance

Metric	Baseline	Our Model	Improvement
MAE	11.65 µg/m^3	**2.78 µg/m^3**	**76.14%**
RMSE	17.74 µg/m^3	**4.31 µg/m^3**	**75.70%**
MAPE	34.74%	**8.92%**	**74.32%**
Coverage	N/A	**84.1%**	Target met

Per City Performance (MAE, µg/m^3)

City	MAE	Note
Karachi	**1.48**	Coastal dispersion advantage
Islamabad	**2.42**	
Lahore	**3.45**	Smog + inversion challenges
Peshawar	**3.77**	

Section 6: Data Pipeline Specs

- **Raw data:** 331,731 PM2.5 + 168,480 weather records
- **Coverage:** Jan 8, 2022 to Oct 20, 2026 (4.8 years)
- **Features:** 27 engineered variables (4 lags, 4 rolling stats, 5 temporal, 8 weather, 6 derived)
- **Model ready:** 76,305 rows x 27 columns
- **Storage:** 20 MB total (Parquet + DuckDB compression)
- **Quality gates:** timezone standardization, deduplication, outlier flagging, range validation

Section 7: Agentic AI Components

- **Data Quality Agent:** autonomous validation with fix, fill, and flag decisions logged to JSONL
- **Forecasting Agent:** hybrid ensemble (LightGBM + CatBoost) with per city regularization
- **RAG Insight Agent:** ChromaDB + sentence-transformers over 7 documents, returning grounded citations (source + page)

Section 8: Deployment and Impact

- **Backend:** FastAPI on Render https://data-insight-api-uybc.onrender.com
- **Frontend:** React on Vercel https://data-insight-two.vercel.app
- **Automation:** Prefect hourly pipelines (5 to 7 min runtime)
- **Impact:** proactive public health interventions 18 to 24h ahead, personalized citizen guidance, policy evaluation with audit trails

Section 9: Future Enhancements

- Expand to 6+ cities (Multan, Faisalabad, Quetta)
- Multi pollutant forecasting (NO_2, SO_2, O_3)
- 48 to 72h horizons with Temporal Fusion Transformer
- Real time SMS and push alerts for unhealthy air
- Mobile app with offline caching
- Satellite data integration (NASA FIRMS, MODIS AOD)

GitHub: https://github.com/Muhammad-Hassan06/Data-Insight | **Live Demo:** https://data-insight-two.vercel.app
API Docs: https://data-insight-api-uybc.onrender.com/docs | **License:** MIT | **Data:** CC-BY-4.0